

Arigala Adarsh

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Education

Rajiv Gandhi University of Knowledge Technologies RK Valley <i>Bachelor of Technology in Computer Science and Engineering</i> • CGPA: 9.1 / 10.00 • Advisor: Prof. Ravi Kumar Penugonda	2021–2025 Andhra Pradesh
Rajiv Gandhi University of Knowledge Technologies RK Valley <i>MBIPC (11th & 12th)</i> • CGPA: 9.85 / 10.00	2019–2021 Andhra Pradesh

Research Areas

Speech and Audio: text-to-speech (TTS), voice conversion (VC), sound event detection (SED) and automatic audio captioning (AAC)

Experience

Project Associate, IIT Madras <i>SPRING Lab, Department of Computer Science and Engineering</i> • Working under Prof. S. Umesh on Text-to-Speech (TTS) and Voice Conversion systems as part of the <i>Speech Technologies in Indian Languages</i> project. • Analyzing and modifying TTS and Voice Conversion architectures for low-resource multilingual adaptation, with a focus on improving duration modeling and alignment stability. • Studying discrete speech units from self-supervised learning (SSL) models for unit-to-speech generation and evaluating their suitability for multilingual speech synthesis.	October 2025 – Present Chennai
Research Intern, IIT Hyderabad <i>Department of Electrical Engineering</i> • Worked under Prof. K. Sri Rama Murty in the Speech Information Processing Lab on the Robot Audition (DRDO) project. • Developed models for audio tagging, sound event classification, and audio search in dynamic environments. • Proposed a unified encoder architecture for reference-guided targeted sound detection in audio mixtures, enabling reliable detection of specific target sounds in complex acoustic environments. • Designed a custom lightweight decoder architecture for audio captioning and incorporated AudioSet-based semantic representations to improve caption generation performance.	May 2024 – Sep 2025 Hyderabad

Projects

Automatic Audio Captioning using PANNs CNN14, ConvNeXt, and BART • Built an Automatic Audio Captioning (AAC) system using PANNs CNN14 and ConvNeXt as audio encoders and BART as the transformer-based decoder. • Processed raw audio into log-mel spectrograms and trained the system on Clotho, AudioCaps, and WaveCaps datasets for diverse caption learning. • Enhanced BLEU score through effective model fine-tuning on multi-dataset training, improving the quality and relevance of generated audio captions.
Sound Event Detection using PANN CNN14 • Fine-tuned the PANN CNN14 model for sound event detection across 18 classes, achieving high accuracy in identifying various audio events. • Integrated the model with Robot Operating System (ROS) for real-time performance, enabling seamless audio classification and interaction in dynamic environments. • Utilized advanced techniques, such as data augmentation and feature extraction, to enhance the model's robustness and performance in diverse acoustic conditions.

Stress Analysis

- Collected data from approximately 800 participants in my surroundings for stress analysis, focusing on identifying factors contributing to stress and understanding stress patterns across different age categories.
- Conducted comprehensive data analysis to identify key stressors and their impact, determining which age groups were most affected and exploring changes in habits and behaviors associated with stress.

Skills

Programming: Python

Tools & Systems: Linux, Git, ROS2

libraries: PyTorch, TensorFlow, NumPy, Pandas, matplotlib, HuggingFace, Librosa

Publications

Conference Publications

- **EZ-AR-VC: Robust Arabic Voice Anonymization with Generalization Across Dialects and Code-Switching**
Injy Hamed, **Arigala Adarsh**, Rufael Marew, Arjun Gangwar, Atharva Kulkarni, Hanan Aldarmaki, Srinivasan Umesh
Submitted to ICASSP2027
- **PIXEL-KWS: Rendered-Glyph Representations for Orthographically Robust Open-Vocabulary Keyword Spotting**
Arigala Adarsh, Ramesh Gundluru, Srinivasan Umesh, Ludovick Lepauloux, Hakim Hacid
Submitted to ICASSP2027
- **Cross-Lingual TTS for Indian Languages: A Benchmark and Inference-Time Duration Estimation Strategies**
Arigala Adarsh*, Arjun Gangwar*, Srinivasan Umesh
Submitted to AACL-IJCNLP 2026 Workshop Multi-LLL
- **Multilingual Multi-Speaker Unit Vocoders: A Systematic Analysis of Discrete Speech Representations**
Naman Kothari, Arjun Gangwar, **Adarsh Arigala**, S. Umesh
Proceedings of Interspeech 2026
- **Reference-Guided Targeted Sound Detection Using a Unified Encoder**
Shubham Gupta*, **Adarsh Arigala***, B. R. Dilleswari, Sri Rama Murty Kodukula
Proceedings of ICASSP 2026)
- **Parasitic Egg Detection and Classification: An Overview of Recent Progress and New Challenges**
D. Gnanavenkata Kumar*, **Adarsh Arigala***, Reddy Palli Trisha, Penugonda Ravikumar
Proceedings of SOMET 2025

Preprints

- **Pixel-TTS: Image Based Text Rendering for Robust Text-to-Speech**
Arigala Adarsh, Arjun Gangwar, Srinivasan Umesh, Yova Kementchedjhieva
- **Enhancing Audio Captioning with Auxiliary AudioSet Semantics**
Shubham Gupta*, **Adarsh Arigala***, Sri Rama Murty Kodukula

*Equal contribution

Volunteering

NSS Member

National Service Scheme (NSS), RGUKT RK Valley

2023 – 2024

Andhra Pradesh

Human Resources Volunteer

Helping Hands Organization, RGUKT RK Valley

2019 – 2025

Andhra Pradesh